Software Requirement Specification

Software Requirement Specification (SRS) — OptiCrop

1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for OptiCrop, a machine learning-based agricultural production optimization engine that recommends suitable crops based on soil and environmental parameters.

1.2 Project Scope

OptiCrop integrates Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH, and rainfall data with trained ML models to provide crop recommendations via a Flask web application. It serves farmers, agricultural researchers, and policymakers.

1.3 Intended Audience

Development team, project mentor/evaluator, and end users (farmers, researchers).

2. Overall Description

2.1 Product Perspective

OptiCrop is a standalone Flask web application backed by a pre-trained ML model (serialized via Pickle) and a soil/crop dataset.

2.2 User Classes

- **Farmer** — enters soil/climate data, receives crop recommendation
- **Researcher/Policymaker** — analyzes crop-environment relationships and trends

2.3 Operating Environment

- OS: Windows / Linux / macOS
- Python 3.10+
- Flask web framework
- Browser-based frontend (HTML, CSS, Bootstrap)

3. Functional Requirements

ID	Requirement
FR-1	System shall accept N, P, K, temperature, humidity, pH, and rainfall as input via a web form
FR-2	System shall validate all input fields before processing
FR-3	System shall preprocess input data (scaling/encoding as required by the model)

FR-4	System shall predict the most suitable crop using the trained ML model
FR-5	System shall display the predicted crop on a results page
FR-6	System shall support assessment of suitability for a specific crop against given conditions
FR-7	System shall allow the underlying model to be retrained/updated with new data

4. Non-Functional Requirements

ID	Requirement
NFR-1	System shall return predictions within 2 seconds of form submission
NFR-2	UI shall be responsive and usable on desktop and mobile browsers
NFR-3	System shall run on minimum hardware: Intel i3, 4GB RAM, 10GB free storage
NFR-4	Trained model shall be persisted (.pkl) to avoid retraining on every prediction
NFR-5	Code shall be modular (separate preprocessing, training, and prediction modules)

5. External Interface Requirements

5.1 User Interface

Web form built with HTML/CSS/Bootstrap, with input fields for N, P, K, temperature, humidity, pH, rainfall, and a “Predict” button.

5.2 Hardware Interfaces

None beyond standard computing hardware; internet connection required for initial dataset/package downloads.

5.3 Software Interfaces

- Flask (web framework)
- Scikit-learn (ML models)
- Pandas / NumPy (data handling)
- Pickle (model persistence)

6. Data Requirements

Dataset containing historical records of: N, P, K, temperature, humidity, pH, rainfall, and corresponding crop label (target variable).

7. Assumptions and Dependencies

- Dataset is reasonably clean and representative of real-world agricultural conditions
- Users have basic internet access to reach the web application
- Model accuracy is dependent on dataset quality and size